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Consumer Product Safety Analysis

1. Business Understanding

1.1 Background

1.1.1 Organizational/Context Description

The Consumer Safety Research Institute (CSRI) is a specialized fictional nonprofit organization established in 2012 to perform independent research on consumer product safety and publish reports and insights that defines public disclosure, and aids in policy decisions. Consumer Safety Research Institute collects comprehensive injury data through systems like National electronic Injury surveillance system (NEISS) – which is a national database managed by U.S Consumer Product Safety Commission (CPSC) to track injuries treated in hospital emergency departments across the United States. While government agencies collect essential data, their raw reports are often cumbersome for the average consumer to understand and comprehend. CSRI fills this critical void, transforming complex datasets into clear public understanding of product risks.

The CSRI publishes an annual "Product Safety Spotlight" report that examines the emerging trends in product safety, identify areas of public attention and provides recommendations for consumers, manufacturers and policymakers. The report serves multiple stakeholders like the parent advocacy groups, policy researchers, product liability attorneys, insurance industry and regulatory agencies.

1.1.2 Domain and Context Assessment

The domain of consumer product safety is currently characterized by a significant disconnect between available data and public

understanding. While the National Electronic Injury Surveillance System (NEISS) provides a robust foundation of comprehensive injury statistics, this raw data is technically complex and largely inaccessible to non-specialists. Consequently, the prevailing public discourse frequently relies on anecdotal evidence, high-profile media incidents, or biased manufacturer marketing rather than systematic, empirical analysis. This reliance on incomplete information leaves consumers, policymakers, and safety advocates without the evidence-based insights necessary for informed decision-making.

This information gap creates a critical challenge in an environment where stakeholders increasingly demand rigorous, independent verification to distinguish between isolated accidents and genuine systemic risks. There is a pressing market need for an authoritative intermediary capable of translating dense surveillance data into clear, actionable intelligence. However, few existing organizations possess the requisite combination of technical data expertise and institutional independence to conduct this analysis effectively.

The Consumer Safety Research Institute (CSRI) is strategically positioned to bridge this divide. By leveraging national injury surveillance data, CSRI transforms abstract statistics into evidence-based answers regarding product safety trends. The institute's analytical work serves to ground public debate in fact, shifting the landscape from reactionary responses toward data-driven policy. In doing so, CSRI not only empowers consumers and regulators with clarity but also proactively identifies emerging safety threats, ensuring that manufacturers and governing bodies can address priority issues with precision and authority.

1.1.3 Problem Situation

While the government collects vast amounts of injury data, there is no effective mechanism to translate this raw information into accessible knowledge for the public.

Without independent analysis, safety decisions are made based on fear, marketing, or anecdotes rather than statistical reality, potentially leaving dangerous products on the market or causing unnecessary panic over minor issues.

There is an urgent requirement for an independent body capable of analyzing complex surveillance data and communicating objective, actionable findings to non-technical stakeholders.

1.1.4 Data Science Justification

Consumer Safety Research Institute (CSRI)has visibility of National Electronic Injury Survey Data (NEISS) data from 2005 through 2024. The availability of NEISS transactional (at injury level) and demographic (Age/Sex//Race) data of 19 years represents a massive, high dimensional repository of public injury information. The volume and complexity of this data make any manual analysis cumbersome and time consuming.

We need to apply advanced machine learning and statistical analysis on this data to gather below insights from this massive dataset.

1.1.4.1 Trend Analysis and Seasonality

1.1.4.2 Identify Unstructured Narrative Data

1.1.4.3 Demographic Clustering

1.1.4.4 Anomaly Detection

1.1.4.5 Identify correlations between predictors causing Injury

1.1.4.6 Complex survey Data and the impact on Statistical Inference.

1.1.4.7 Identify predictors causing hospitalization due to injuries.

1.1.5 Stakeholder Identification

The CSRI organization helps safety experts understand the emerging hazards caused due to consumer product usage. They use this knowledge to verify the danger and protect public from injuries related to consumer products.

The report generated by CSRI also aids the safety and product engineers perform an in-depth analysis of the root cause for the injuries and then perform a risk assessment where they compare the severity of the injury to the probability of the risk and then identifies the vulnerable population.

The reports and insights generated by CSRI is also used by experts and government agencies to issue product recalls and to rewrite the safety rules that the manufactures need to follow. The knowledge gained from these reports are also broadcasted for public disclosure to prevent future injuries in the form of safety alerts.

The findings in these reports are also published to warn manufacturers, retailers, insurers and policy makers that a new regulation is needed to prevent injuries.

1.2 Project Objectives and Success Criteria

1.2.1 Primary Objective

Develop predictive models to identify which consumer product injuries are most likely to result in hospitalization and determine what factors—product categories, demographics, injury characteristics—predict severe medical outcomes among emergency department cases, producing evidence-based findings that inform CSRI's 2025

Product Safety Spotlight report and contribute to public understanding of consumer product safety priorities.

1.2.2 Secondary Objectives

Identify temporal and demographic trends in consumer product injury severity patterns, including seasonal variations, changes over time, and emerging patterns that may indicate shifting safety concerns, generate insights about demographic patterns in severe product injuries that inform targeted safety messaging, and create analytical approaches that enhance CSRI's reputation as a leading source of independent consumer safety research.

1.2.3 Quantitative Success Criteria

1.2.3.1 Model Performance

Predictive Accuracy: The predictive models should achieve a Receiver Operating Characteristic (ROC) AUC score of > 0.70 thereby indicating an ability to strongly distinguish between severe (hospitalized) vs non-severe cases (not hospitalized).

Statistical Significance: The predictors identified from the statistical analysis must demonstrate statistical significance with $p < 0.05$ to be considered valid predictors.

Goodness of Fit: The model must pass standard calibration tests to ensure that predicted probabilities reasonability match observed hospitalization.

1.2.3.2 Trend Analysis

The analysis must successfully quantify seasonal variations (identify peak injury months) and long-term trends (annualized percentage change). The analysis should also identify at least 3-5 emerging injury patterns that signals shifting safety concerns.

1.2.3.3 Output Deliverables

The project must generate at least 3-5 visualizations and 3 summary tables for the inclusion in the CSRI 2025 Safety Spotlight Report

1.2.4 Qualitative Success Criteria

1.2.4.1 Actionability of Insights

The insights gathered from the statistical analysis should be specific enough to be translated into actionable safety advice.

1.2.4.2 Model Interpretability and Explainability

The factors causing hospitalization should be explainable to non-technical audiences and should not be a black box.

1.2.4.4 Organizational Impact

The analytical approach should be robust enough to withstand the scrutiny of external reviewers and industry critics thereby enhancing CSRI's reputation.

The analytical approaches should be documented and reproducible.

The work must meet the institute's standards for scientific rigor and public communication.

The analysis will be published under CSRI's name and must withstand scrutiny from diverse stakeholders including consumer advocates, industry representatives, academic researchers, and policy makers.

1.2.5 Impact Assessment

The report will provide Consumer advocacy groups and policymakers data-backed evidence to support safety recommendations and regulatory decisions.

By systematically analyzing injury patterns, the report will contribute to more informed public discourse while identifying emerging safety issues that warrant attention from consumers, manufacturers, and regulators.

1.2.6 Timeline and Milestones

Expect to complete the statistical analysis within 8 weeks, draft findings within 10 weeks and publish the final article within 12 weeks of project start.

1.3 Assessment of Responsibility

1.3.1 Legal and Regulatory Considerations

The data that is available through NEISS is a public domain information and will not need any copyrights to use the information.

This data does not have any Personally Identifiable information like the name, address, SSN, Manufacturer or Hospital information.

We should not make any attempts during /after the project life cycle to re-identify the persons, hospitals, manufactures by linking this data to other external datasets/vectors.

CPSC is bound by the Consumer Product Safety Act (CPSA) Section 6(b) and Privacy Act of 1974 that strictly regulates the release of information that could identify individuals or manufacturers without a fair chance to respond.

CPSC requires that any public analysis of the data acknowledges the source and include a disclaimer "This analysis is based on data from the U.S. Consumer Product Safety Commission (CPSC) but is not an official determination or policy of the CPSC." (Needs to be added in the Phase-6 Deployment Section)

1.3.2 Privacy and Data Protection

Patient Anonymity:

The NEISS data is available for public consumption and is deidentified to remove any PII comply with the Privacy Act of 1974 and HIPPA Act.

The data is devoid of any direct identifiers (Name, Address, Phone Numbers, SSN) before released. The Treatment Date in the dataset is preserved to maintain the seasonality but the time in the data omitted or generalized/shifted to prevent direct/indirect identification from the police report or ambulance log. The narratives are also scrubbed to remove any PII information from creeping in. We should not make any attempts during /after the project life cycle to re-identify the persons.

Hospital Confidentiality:
The identities of 100 participating hospitals are kept confidential, but NEISS releases a geographical map which helps us to identify were/which state and the strata of the hospitals participating the program. No hospital analysis is possible using the NEISS dataset for this project.

Manufacturer Protection:
The consumer product safety act (Section 6(b)) restricts CPSC from identifying any manufacturers in the data. Brand/Manufacturer information in not included in the NEISS dataset and hence we need to summarize the analysis based on product categories.

Statistical Disclosure Limitation:
Rare events or data could help in re-identification like (110-year-old male injured by a product on a specific day). CPSC rule needs us to follow cell suppression rules if a query yields fewer than 20 cases nationwide.

1.3.3 Bias and Fairness Concerns

1.3.3.1 Selection Bias

The NEISS dataset includes data for injuries that were only reported to the Emergency Departments and does not include the information of people going to the primary care physicians, urgent care or individuals taking self-care. This data also does not include information of people who died at the scene due to a product usage as they are taken directly to the morgue.

This introduces fairness concerns as lower income people might not have a PCP and could always use the ED for non-emergencies. This could inflate the injury rates for people with lower income while undercounting injuries of people with higher income.

1.3.3.2 Reporting Bias

The data is manually entered into the CPSC system by hospital NEISS hospital coordinators using the data entered by doctors and nurses into the patients' medical record. This data could have been entered hurriedly or could be incomplete, and this could possibly introduce reporting bias whereby the data could have many unknown values. If the model drops rows with NULL/UNKNOWN data, we could be possibly excluding data that could have been useful.

1.3.3.3 Socio Economic Bias

Exposure to injury also depends on socio-economic background, as the exposure to products is not random. If the analysis only focuses on the number of injuries without adjusting for the exposure (number of people who owning the product) then it could cause a bias.

1.3.3.4 Language Bias

The narratives against the NEISS case are written in English. If the patient doesn't know English, they might not be able to effectively

communicate how injury happened to the primary care givers or to the nurse/doctors.

1.3.3.5 Geographic Bias

The NEISS data is not collected for every hospital in the USA. NEISS is a probability sample, and it generalized the hospitals into 100 buckets. This could potentially mask localized issues.

1.3.5 Stakeholder Impact Analysis

Positive impacts include informed residents making better product purchasing decisions and evidence-based policy making; Design changes made by the manufactures to reduce lawsuits and increase in profitability; Insurers reducing product liability premiums for manufactures.

1.3.5 Mitigation Strategies

Use statistical methods that account for monitoring bias; present findings with appropriate uncertainty ranges; include community context in reporting; establish fact-checking protocols for public-facing analysis.

1.4 Data Science Goals & Success Criteria

1.4.1 Technical Problem Framing

Frame as a classification problem to predict the relationship between factors affecting physical injuries and subsequent hospitalization due to the use of consumer products. The analysis should also clearly identify the patterns of injury severity across product categories and demographic groups. The analysis should also contribute to evidence-based safety recommendations.

1.4.2 Data science Objectives

Primary data science goal is to establish statistically significant correlations between factors contributing to physical injuries and hospitalization due to the use of consumer products with a high degree of confidence; secondary goal is to create interpretable models that quantify effect sizes for public communication

1.4.3 Technical Success Criteria

1.4.3.1 Development of statistically sound predictive models that can reliably identify factors associated with hospitalization for product-related injuries.

1.4.3.2 Clear identification of meaningful patterns in injury severity across different product categories and demographic groups.

1.4.3.3 Rigorous analysis that produces findings robust enough to support evidence-based safety recommendations.

1.4.3.4 Effectively situating insights from the analysis to contribute new understanding to consumer product safety discourse and inform practical safety guidance.

1.4.3.5 Effective communication of technical findings to diverse audiences including consumers, policymakers, and safety advocates through accessible analysis and clear visualizations.

1.4.4 Analytical Approach Overview

Plan to compare logistic regression (base model), decision tree, random forest, gradient boosting (XGBoost) to predict hospitalization;

use K-means and hierarchical clustering for customer segmentation; evaluate using cross-validation and business-relevant metrics.

1.4.5 Objective-Technical Mapping

Statistically significant correlation analysis with quantified effect sizes enables editorial team to make evidence-based claims about the factors impacting physical injuries due to product usage that can cause severe trauma needing hospitalization, supporting the goal of providing residents with credible information for personal decision-making.

1.4.6. Constraints and Assumptions

Assume customer behavior patterns remain relatively stable over the analysis period; model performance may be limited by data quality issues identified in preliminary exploration; computational constraints may require sampling for initial model development.

1.5 Project Plan

1.5.1 Phase specific planning

Phase-1 Business Understanding

1. Determine Business Objectives
   Activity: Confirm the primary goal – Identify the factors predicting the hospitalization
2. Assess Situation
   Activity: Inventory of NEISS Files and Coding Manual
3. Determine Data Mining Goals
   Activity: Set success metric and define the binary target variable (based on disposition)
4. Produce Project Plan
   Activity: Outline Resources, timeline and tools

Phase-2 Data Understanding

1. Collect Initial Data
   Activity: Ingest raw files into a project environment
2. Describe Data
   Activity: Map schema and confirm the relationships
3. Explore Data

Activity: Generate histograms and generate EDA report

4. Verify Data Quality
   Activity: Check for missing or incorrect data

Phase-3 Data Preparation

1. Select Data
   Activity: Filter out of scope data records from the dataset
2. Clean Data
   Activity: Clean data if needed
3. Construct Data
   Activity: Perform Feature Engineering
4. Integrate Data
   Activity: Integrate transactional data with other data dictionaries to increase the readability
5. Format Data
   Activity: Perform encoding and finalize data for modelling

Phase-4 Modelling

1. Select Modelling Techniques
   Activity: Select Base model and possible modelling algorithms.
2. Generate Test Design
   Activity: Split data into Training (70%), Test (30%) – stratified by disposition.
3. Build Model
   Activity: Train Models to predict Hospitalization
4. Assess Model
   Activity: Verify the performance of the model (ROC/AUC, Confusion Matrix)

Phase-5 Evaluation

1. Evaluate Results
   Activity: Review results and confirm if it meets goodness of fit.
2. Review Process
   Activity: Ensure that small samples (N<20) are not used in the release
3. Determine Next Steps
   Activity: Determine if the findings are robust enough

Phase-6 Deployment

1. Plan Deployment
   Activity: Create a deployment plan
2. Produce Final Report
   Activity: Draft narrative to explain the findings

3. Final Presentation
   Activity: Present findings to stakeholders
4. Review Project
   Activity: Document lessons learned and code documentation.

1.5.2 Timeline and milestones

Phase-1 Business Understanding (1-2 Weeks)

Phase-2 Data Understanding (1 Week)

Phase-3 Data Preparation and Initial EDA (Week 4)

Phase-4 Modelling – Finish baseline models (Week 6)

Phase-5 Evaluation – Complete model evaluation and selection (Week 8)

Phase-6 Deployment - Finalize recommendations for the project (Week 10)

1.5.3 Risk identification and contingencies

1.5.3.1 Data Quality Risks

Risk: Large number of records with missing data

Contingency: Imputation and Segmentation – Do not delete the rows with missing or unknown values but treat these as another segment/category. Another option is to use the mode for categorical variable and median if a numeric feature rather than amputation of the rows.

Risk: Ambiguous Narrative – The data in the narrative column could be ambiguous and might not offer much value.

Contingency: Use stop words and If narrative is too vague then exclude.

1.5.3.2 Technical Risks

Risk: Model performance may not meet targets

Contingency: Use a baseline model and multiple models and then evaluate the performance.

Risk: Lack of Model Interpretability

Contingency: Train on simple base line model (logistic regression) alongside complex models.

Risk: Model Overfitting

Contingency: Use Cross Validation (Use time series splits train on the past and test on the future is an option)

1.5.3.3 Legal Risks

Risk: Violation of Section 6(b)

Contingency: Ensure that no brand names from the narrative's percolates into the report.

Risk: Small sample stability

Contingency: Use suppression rules to suppress any estimates based on fewer than 20 sample size.

1.5.3.4 Operational Risks

Risk: Seasonal Confounding – Confusing a seasonal spike with genuine product defect

Contingency: Apply statistical "Seasonal Decomposition" to segregate Trend/Seasonality/Noise before confirming a spike

2 Data Understanding

2.1 Data Inventory

2.1.1 Primary Dataset Description

The primary dataset used by the Consumer Safety Research Institute (CSRI) is the National Electronic Injury Surveillance System (NEISS). Managed by the U.S. Consumer Product Safety Commission (CPSC), this database serves as a nationally representative probability sample of hospitals in the United States and its territories. NEISS injury data are gathered from the emergency departments (ED) of approximately 100 hospitals selected as a probability sample of all 5,000+ U.S. hospitals with emergency departments. The system's foundation rests on emergency department surveillance data, but the system also has the flexibility to gather additional data at either the surveillance or the investigation level.

Scope of Data

NEISS consumer product related data is available online for the last 20 years. The most recent calendar year of treatment dates generally become available in April of the following year.

Source of Data

URL: https://www.cpsc.gov/cgibin/NEISSQuery/home.aspx

The NEISS Data is collected by the Consumer Product Safety Commission and shared to the public via their website.

NEISS Overview page: https://www.cpsc.gov/Research--Statistics/NEISS-Injury-Data

NEISS FAQ page: https://www.cpsc.gov/Research--Statistics/NEISS-Injury-Data/Neiss-Frequently-Asked-Questions

How is data stored and downloaded?

We can download data from the NEISS portal either as an excel or as a tab limited text or a SAS format. The data can be downloaded from the portal by specifying the data year.

The files downloaded from NEISS, will have the name in the format neissYYYY.xlsx. Each excel file will have data for the year that we downloaded. Below is the list of files that we downloaded from NEISS portal for the year 2005-2024.

['neiss2005.xlsx', 'neiss2006.xlsx', 'neiss2007.xlsx', 'neiss2008.xlsx', 'neiss2009.xlsx', 'neiss2010.xlsx', 'neiss2011.xlsx', 'neiss2012.xlsx', 'neiss2013.xlsx', 'neiss2014.xlsx', 'neiss2015.xlsx', 'neiss2016.xlsx', 'neiss2017.xlsx', 'neiss2018.xlsx', neiss2019.xlsx', 'neiss2020.xlsx', 'neiss2021.xlsx', 'neiss2022.xlsx', 'neiss2023.xlsx', 'neiss2024.xlsx']

2.1.2 Additional Data Sources

None

2.1.3 Data Collection Methodology

NEISS injury data are gathered from the emergency departments (ED) of approximately 100 hospitals selected as a probability sample of all 5,000+ U.S. hospitals with emergency departments. The system's foundation rests on emergency department surveillance data, but the system also has the flexibility to gather additional data at either the surveillance or the investigation level. Surveillance data enable CPSC analysts to make timely national estimates of the number of injuries associated with (but not necessarily caused by) specific consumer products. This data also provides evidence of the need for further study of products. Subsequent follow-back studies yield important clues to the cause and likely prevention of injuries and deaths.

The NEISS data collection methodology initiates at the hospital level, where emergency department staff record injury details during patient admission. A designated NEISS hospital coordinator reviews these daily records to identify cases meeting specific inclusion criteria. The coordinator abstracts pertinent information, translating it into standardized codes based on the NEISS Coding Manual. This coding process captures critical variables, including patient demographics (age, race, ethnicity), injury diagnosis, incident locale, and highly specific product codes that distinguish between similar items (e.g., different types of lawn mowers). A brief narrative description is also added to provide context not captured by codes.

Data entry is performed using CPSC-designed software that applies real-time validation logic to reject inconsistent entries, such as anatomically impossible injuries. Once validated locally, the data is transmitted via a secure internet connection to the CPSC. Upon receipt, the data undergoes a secondary automated editing process before being integrated into the permanent NEISS database. This streamlined workflow allows CPSC analysts in the Directorate for Epidemiology to review the data on the same day it is collected, facilitating immediate quality control and the rapid detection of emerging consumer safety hazards.

2.2 Data Description

2.2.1 Comprehensive Data Dictionary

NEISS provides us two data dictionaries, one for data collected prior to 2018 and another for data collected beyond 2019. Below you can find both the data dictionaries.

Further information about the dataset is available in the 2025 NEISS Coding Manual available online.

Prior to 2018:

Excel/Tab Variable	SAS Variable	Description/Code book
CPSC_Case_Number	NEK	CPSC case number
Treatment_Date	Trmt_date	Date of Treatment
Age	Age	Age of Patient
		0 - Age Not Recorded
		2 to 120 - Age in Years
		201 to 223 - Age 1 to 23 Months
Sex	Sex	Sex of Patient
		0 - Not Recorded
		1 - Male
		2 - Female
Race	Race	Race of Patient
		0 = Not Stated in ED Record
		1 = White
		2 = Black/African American
		3 = Other
		4 = Asian
		5 = American Indian/Alaska Native
		6 = Native Hawaiian/Pacific Islander
Other_Race	Raceoth	Description of Other Race (Used with Race=3)
Body_Part	Bdpt	Injured Body Part
		0 - Internal
		30 - Shoulder
		31 - Upper Trunk
		32 - Elbow
		33 - Lower Arm
		34 - Wrist
		35 - Knee
		36 - Lower Leg
		37 - Ankle
		38 - Pubic Region
		75 - Head
		76 - Face
		77 - Eyeball
		78 - Upper Trunk(Old)
		79 - Lower Trunk
		80 - Upper Arm
		81 - Upper Leg
		82 - Hand
		83 - Foot
		84 - 25-50% of Body
		85 - All Parts Body

		86 - Other(Old)
		87 - Not Stated/Unk
		88 - Mouth
		89 - Neck
		90 - Lower Arm(Old)
		91 - Lower Leg(Old)
		92 - Finger
		93 - Toe
		94 - Ear
Diagnosis	Diag	Injury Diagnosis
		41 - Ingestion
		42 - Aspiration
		46 - Burns, Electrical
		47 - Burns, Not Specified
		48 - Burns, Scald
		49 - Burns, Chemical
		50 - Amputaion
		51 - Burns, Thermal
		52 - Concussions
		53 - Contusions, Abrasions
		54 - Crushing
		55 - Dislocation
		56 - Foreign Body
		57 - Fracture
		58 - Hematoma
		59 - Laceration
		60 - Dental Injury
		61 - Nerve Damage
		62 - Internal Organ Injury
		63 - Puncture
		64 - Strain, Sprain
		65 - Anoxia
		66 - Hemorrhage
		67 - Electric Shock
		68 - Poisoning
		69 - Submersion
		71 - Other/Not Stated
		72 - Avulsion
		73 - Burns, Radiation
		74 - Dermatitis, Conjunctivitis
Other_Diagnosis	Diagoth	Description of Other Diagnosis (Used with Diag=71)
Disposition	Disp	Disposition
		1 - Treated/Examined and Released
		2 - Treated and Transferred
		4 - Treated and Admitted/Hospitalized

		5 - Held for Observation
		6 - Left Without Being Seen
		8 - Fatality, Incl. DOA, Died in ER
		9 - Unknown, Not Stated
Location	Loc	Incident Location
		0 - Not Recorded
		1 - Home
		2 - Farm/Ranch
		4 - Street or Highway
		5 - Other Public Property
		6- Mobile/Manufactured Home
		7 - Industrial
		8 - School/Daycare
		9 - Place of Recreation or Sports
Fire_Involvement	Fmv	Fire Involvement
		0 - No Fire Involved or Fire Involvement Not Recorded
		1 - Fire Involved and Fire Department Attended
		2 - Fire Involved and Fire Department Did Not Attend
		3 - Fire Involved and Unknown Fire Department Attendance
Product_1	Prod1	Product Code (See NEISS Coding Manual for Details)
Product_2	Prod2	Product Code 2 (See NEISS Coding Manual for Details)
Narrative	Narr	Description of Injury Event
Stratum	Stratum	Design Variable-Stratum
PSU	PSU	Design Variable-Primary Sampling Unit (PSU)
Weight	Wt	Statistical Weight for National Estimates

2019 and Beyond:

Excel/Tab Variable	SAS Variable	Description/Code book
CPSC_Case_Number	NEK	CPSC case number
Treatment_Date	Trmt_date	Date of Treatment
Age	Age	Age of Patient
		0 - Age Not Recorded
		2 to 120 - Age in Years
		201 to 223 - Age 1 to 23 Months
Sex	Sex	Sex of Patient
		0 - Not Recorded

		1 - Male
		2 - Female
		3 - Non-Binary/Other (NA before 2021)
Race	Race	Race of Patient
		0 = Not Stated in ED Record
		1 = White
		2 = Black/African American
		3 = Other
		4 = Asian
		5 = American Indian/Alaska Native
		6 = Native Hawaiian/Pacific Islander
Other_Race	Raceoth	Description of Other Race (Used with Race=3)
Hispanic	Hispanic	Hispanic, Latino/Latina, or of Spanish Origin - available from 2019
		0 - Unknown
		1 - Yes
		2 - No
Body_Part	Bdpt	Injured Body Part
		0 - Internal
		30 - Shoulder
		31 - Upper Trunk
		32 - Elbow
		33 - Lower Arm
		34 - Wrist
		35 - Knee
		36 - Lower Leg
		37 - Ankle
		38 - Pubic Region
		75 - Head
		76 - Face
		77 - Eyeball
		79 - Lower Trunk
		80 - Upper Arm
		81 - Upper Leg
		82 - Hand
		83 - Foot
		84 - 25-50% of Body
		85 - All Parts Body
		87 - Not Stated/Unk
		88 - Mouth
		89 - Neck
		92 - Finger

		93 - Toe
		94 - Ear
Body_Part_2	Bdpt2	Injured Body Part 2- available from 2019
Diagnosis	Diag	Injury Diagnosis
		41 - Ingestion
		42 - Aspiration
		46 - Burns, Electrical
		47 - Burns, Not Specified
		48 - Burns, Scald
		49 - Burns, Chemical
		50 - Amputaion
		51 - Burns, Thermal
		52 - Concussions
		53 - Contusions, Abrasions
		54 - Crushing
		55 - Dislocation
		56 - Foreign Body
		57 - Fracture
		58 - Hematoma
		59 - Laceration
		60 - Dental Injury
		61 - Nerve Damage
		62 - Internal Organ Injury
		63 - Puncture
		64 - Strain, Sprain
		65 - Anoxia
		66 - Hemorrhage
		67 - Electric Shock
		68 - Poisoning
		69 - Submersion
		71 - Other/Not Stated
		72 - Avulsion
		73 - Burns, Radiation
		74 - Dermatitis, Conjunctivitis
Diagnosis_2	Diag2	Injury Diagnosis 2 - available from 2019
Other_Diagnosis	Diagoth	Description of Other Diagnosis (Used with Diag=71)
Other_Diagnosis_2	Diagoth2	Description of Other Diagnosis 2 (Used with Diag2=71) - available from 2019
Disposition	Disp	Disposition
		1 - Treated/Examined and Released
		2 - Treated and Transferred
		4 - Treated and Admitted/Hospitalized

		5 - Held for Observation
		6 - Left Without Being Seen
		8 - Fatality, Incl. DOA, Died in ER
		9 - Unknown, Not Stated
Location	Loc	Incident Location
		0 - Not Recorded
		1 - Home
		2 - Farm/Ranch
		4 - Street or Highway
		5 - Other Public Property
		6- Mobile/Manufactured Home
		7 - Industrial
		8 - School/Daycare
		9 - Place of Recreation or Sports
Fire_Involvement	Fmv	Fire Involvement
		0 - No Fire Involved or Fire Involvement Not Recorded
		1 - Fire Involved and Fire Department Attended
		2 - Fire Involved and Fire Department Did Not Attend
		3 - Fire Involved and Unknown Fire Department Attendance
Alcohol	Alcohol	Consumed Alcohol Prior to or During the Incident - available from 2019
		0 - No/No Information
		1 - Yes
Drug	Drug	Use of Drugs or Medication Contributed to the Incident or the Severity of the Injury - available from 2019
		0 - No/No information
		1 - Yes
Product_1	Prod1	Product Code (See NEISS Coding Manual for Details)
Product_2	Prod2	Product Code 2 (See NEISS Coding Manual for Details)
Product_3	Prod3	Product Code 3 (See NEISS Coding Manual for Details) - available from 2019
Narrative	Narr	Description of Injury Event
Stratum	Stratum	Design Variable-Stratum
PSU	PSU	Design Variable-Primary Sampling Unit (PSU)
Weight	Wt	Statistical Weight for National Estimates

2.2.2 Dataset Structure and Description

The dataset provided by NEISS is denormalized conceptually into a star schema with a centralized fact table with injury information and several dimension tables describing different attributes.

2.2.2.1 Fact Table (Transactional Data)

The primary dataset is a file with injury data that is transactional. Each row in the main table represents a unique injury that is getting treated at a hospital. Each record represents injury at a patient level.

Key Information:

Case Number (Identifier-Primary key)

Treatment Date – Date

Demographics (Age, Sex, Race, Other Race) – Coded Integers

Injury Information (Diagnosis, Other Diagnosis, Body Part, Disposition)

Location – Coded Integer

Product Information (Product_1, Product_2, Product_3) – Coded Integers

Narrative Data – Unstructured Data

Fire Involvement – Coded Integer

Statistical Weight – Numeric Value

2.2.2.2 Dimension Tables (Relationships)

The entire NEISS data is present in a denormalized form in an excel file, but the data relies on the logic connecting the NEISS coding manual. The NEISS dataset is encoded and the integer values for the below fields are linked to corresponding description (meaning) from the dimension tables.

Code to Description Relationship

Race: Race is encoded and represents the patients race in the ED record.

Gender: Reports patients gender as in the ED record.

Hispanic: Hispanic variable indicates the ethnicity.

Product Code (Product 1, Product 2, Product 3): A product code is uniquely mapped to the product description. If more than one product is involved in the injury, the data is available in the product 1, product 2 or product 3 features.

Body Part (Body Part 1 and Body part 2): The body part code is mapped to the body part description. If more than one body part is involved in the injury, corresponding data is available in the body_part_2 feature.

Diagnosis (Diagnosis 1 and Diagnosis 2): The diagnosis code is uniquely mapped to the diagnosis description. If more than one diagnosis is found, the corresponding information is available in the diagnosis_2 feature.

Disposition: Disposition code reports the patient's final disposition as recorded in the ED record.

Location: Location code denotes the location where the injury occurred.

Fire Involvement: Fire involvement code defines if a fire was involved in the injury.

2.2.3 Temporal Coverage and Granularity

The data spans from January 2005 to December 2024 with daily injury records reported to the ED. The coverage of data is 100% across all calendar days of the month between January 2005 to December 2024. The number of injuries seems to have spiked in 2010. We can see a dip in the number of injuries between 2019-2020 due to the COVID19 pandemic and then stabilizing to the pre-pandemic levels.

2.2.4 Data Volume and Scale

Complete dataset contains 7,315,732 transactions with 23 features across 839 products; raw data files total 896.4 MB uncompressed; processing requires approximately 4GB RAM for in-memory analysis.

2.2.5 Categorical Variable Encoding

The below categorical variables are encoded in the NEISS dataset

2.2.5.1 Sex

Recorded Gender in ED Record	Code
Male	1
Female	2
Gender diverse & intersex encompasses but is not limited to all of the following identities: transgender, male to female; transgender, female to male; transgender, gender nonconforming; gender fluid; genderqueer; polygender/pangender; omnigender; non-binary; bigender; agender; Two-spirit identities; Fa'afafine; Hijras; Khanith; intersex.	3
Not recorded/Unknown	0

2.2.5.2 Race/Other Race

Definition	Race Code
A person having origins in any of the original peoples of Europe, the Middle East, or North Africa.	1-White
A person having origins in any of the black racial groups of Africa.	2-Black/African American
Use this code when: (1) the ED record indicates more than one race (*e.g.*, multiracial, biracial), or (2) the race is stated in the ED record and none of the above applies.	3-Other (then enter race specified by ED record in the **Race Other** field)
A person having origins in any of the original peoples of the Far East, Southeast Asia, or the Indian subcontinent.	4-Asian
A person having origins in any of the original peoples of North and South America (including Central America), and who maintains tribal affiliation or community attachment.	5-American Indian/Alaska Native
A person having origins in any of the original peoples of Hawaii, Guam, Samoa, or other Pacific Islands.	6-Native Hawaiian/Pacific Islander
Use this code when there is no information in the ED record.	0-Not stated

2.2.5.3 Hispanic

If the ED Record …	Then Code
indicates the patient **is** a person of Hispanic, Latino/Latina, or Spanish origin.	1-Yes
indicates the patient **is not** a person of Hispanic, Latino/Latina, or Spanish origin.	2-No
does not indicate whether the patient is of Hispanic, Latino/Latina, or Spanish origin.	0-Unknown

2.2.5.4 Body Part/Body Part 2

Body Part Affected	Code
Arm, lower (not including elbow or wrist)	33
Arm, upper	80
Ankle	37
Ear	94
Elbow	32
Eyeball	77
Face (including eyelid, eye area and nose)	76
Finger	92
Foot	83
Hand	82
Head	75
Internal (use with aspiration and ingestion)	0
Knee	35
Leg, lower (not including knee or ankle)	36
Leg, upper	81
Mouth (including lips, tongue and teeth)	88
Neck	89
Pubic region	38
Shoulder (including clavicle, collarbone)	30
Toe	93
Trunk, lower	79
Trunk, upper (not including shoulders)	31
Wrist	34
25-50% of body (used for burns only)	84
All parts of body (more than 50% of body)	85
Not recorded	87

2.2.5.5 Diagnosis/Diagnosis 2

Diagnosis	Code
Amputation	50
Anoxia	65
Aspirated foreign object	42
Avulsion	72
Burns, scald (from hot liquids or steam)	48
Burns, thermal (from flames or hot surface)	51
Burns, chemical (caustics, *etc.*)	49
Burns, radiation (includes all cell damage by ultraviolet, x-rays, microwaves, laser beam, radioactive materials, *etc.*)	73
Burns, electrical	46
Burns, not specified	47
Concussions	52
Contusions, Abrasions	53
Crushing	54
Dental injury	60
Dermatitis, Conjunctivitis	74
Dislocation	55
Electric shock	67
Foreign body	56
Fracture	57
Hematoma	58
Hemorrhage	66
Ingested foreign object	41
Internal organ injury	62
Laceration	59
Nerve damage	61
Poisoning	68
Puncture	63
Strain or Sprain	64
Submersion (including Drowning)	69
Other/Not Stated	71*

2.2.5.6 Disposition

Disposition	Code
Treated and released, or examined and released without treatment (also includes transfers for treatment to another department of the same facility without admission)	1
Treated and transferred to another hospital	2
Treated and admitted for hospitalization (within same facility)	4
Held for observation (includes admitted for observation)	5
Left without being seen, Left against medical advice, Left without treatment, Eloped	6
Fatality, includes dead on arrival ("DOA"), died in the ED, and died after admission	8
Not recorded	9

2.2.5.7 Location

Location	Code
Unknown or not recorded	0
Home	1
Farm/Ranch	2
Street or highway	4
Other public property	5
Manufactured (Mobile) home	6
Industrial place	7
School	8
Place of recreation or sports	9

2.2.5.8 Fire Involvement

Fire Involvement	Code
No fire or no flame/smoke spread	0
Fire involvement and/or smoke inhalation - Fire Department ***attended***	1
Fire involvement and/or smoke inhalation - Fire Department ***did not attend***	2
Fire involvement and/or smoke inhalation - Fire Department ***attendance is not recorded***	3

2.2.5.9 Product 1/Product 2/Product 3

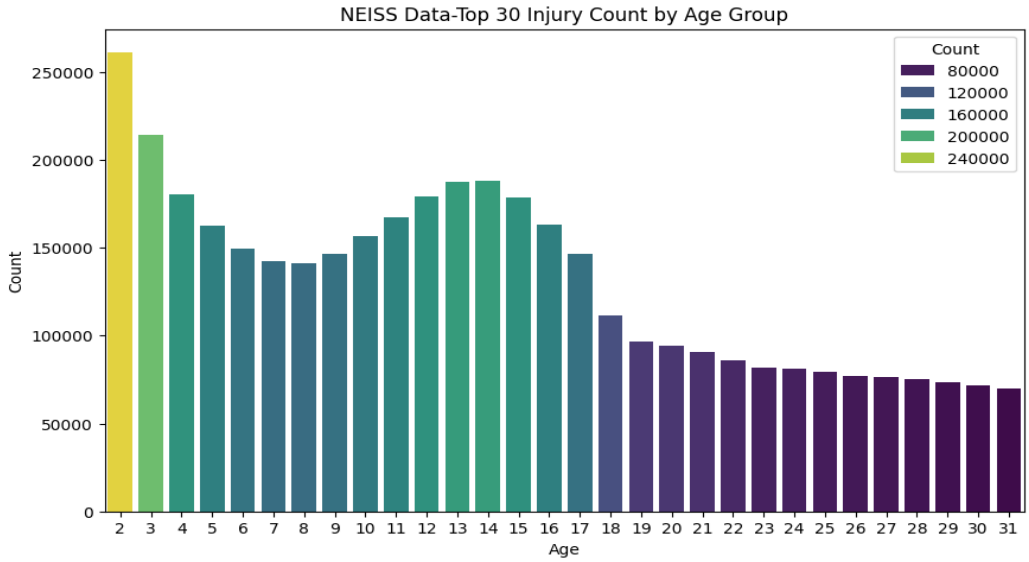
Refer the appendix section of the NEISS Coding manual.

https://www.cpsc.gov/cgibin/NEISSQuery/Data/Info%20Docs/2023%20NEISS%20Coding%20Manual.pdf

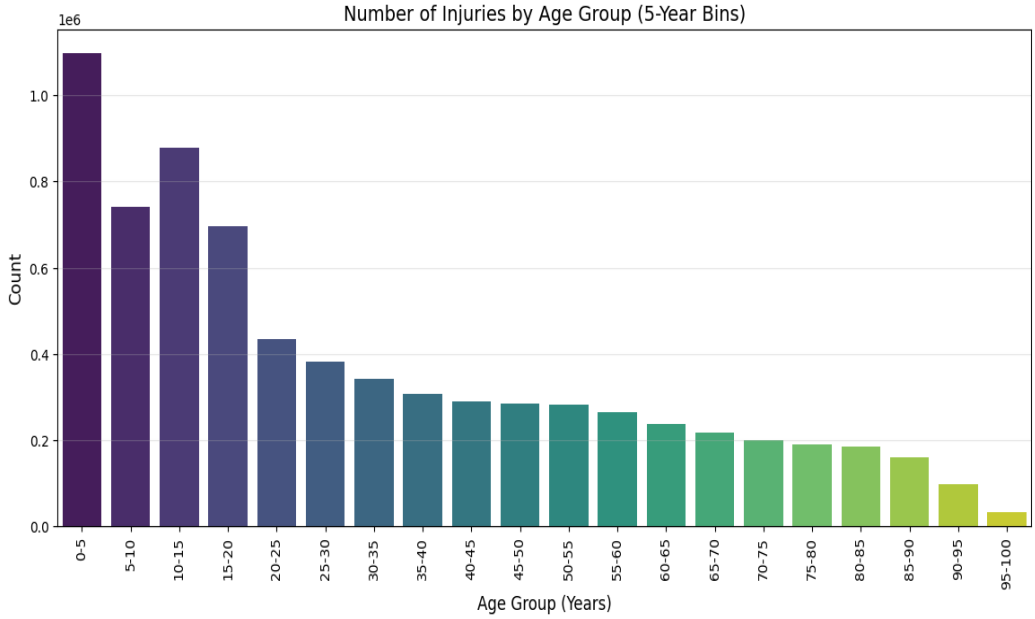
2.3 Data Exploration Report

2.3.1 Univariate Analysis

Top 30 Injury count by Age group

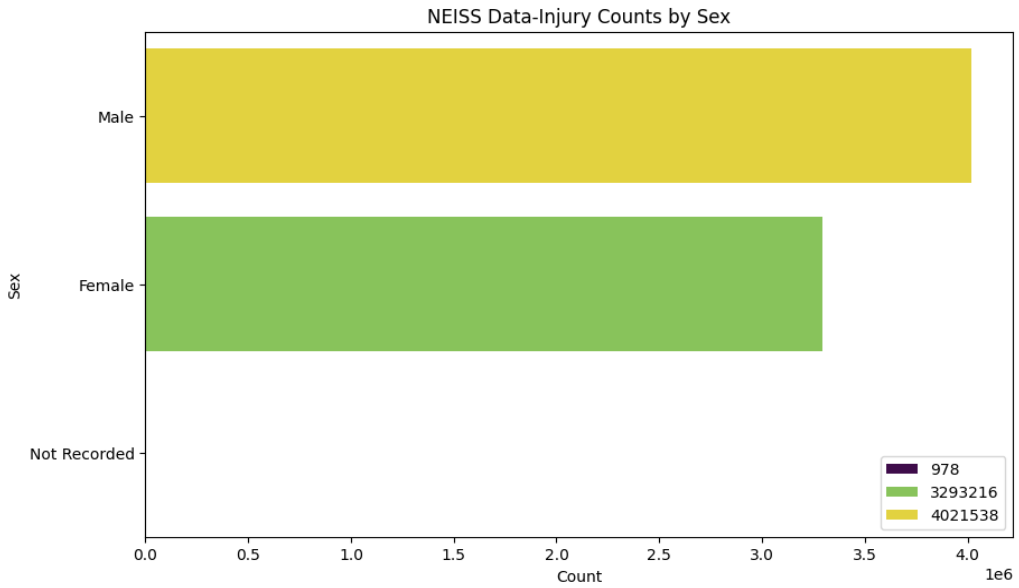


From the chart, we can see that the injury count is high among children and then after the age of 18, stabilizes. The number of injuries seems to peak for children of age 1-2 years.



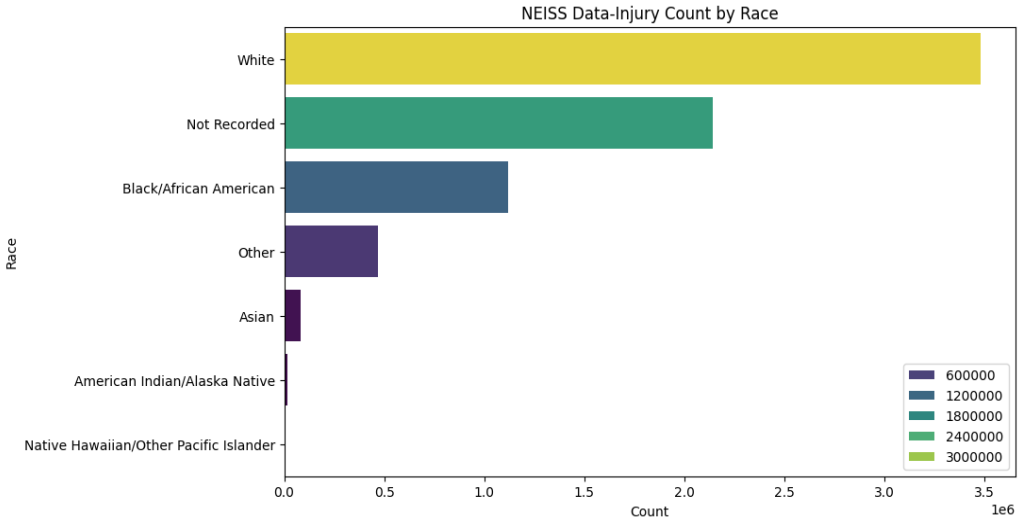
The number of injuries reduces after the age of 75 years.

Injury Count by Sex



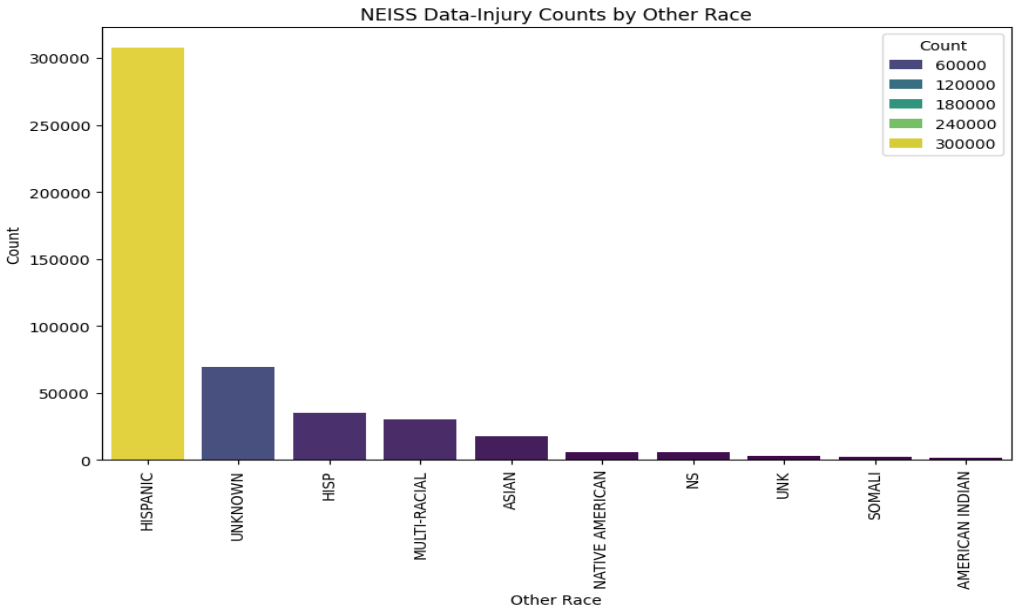
The injury due to the use of consumer products is more among males.

Injury Count by Race



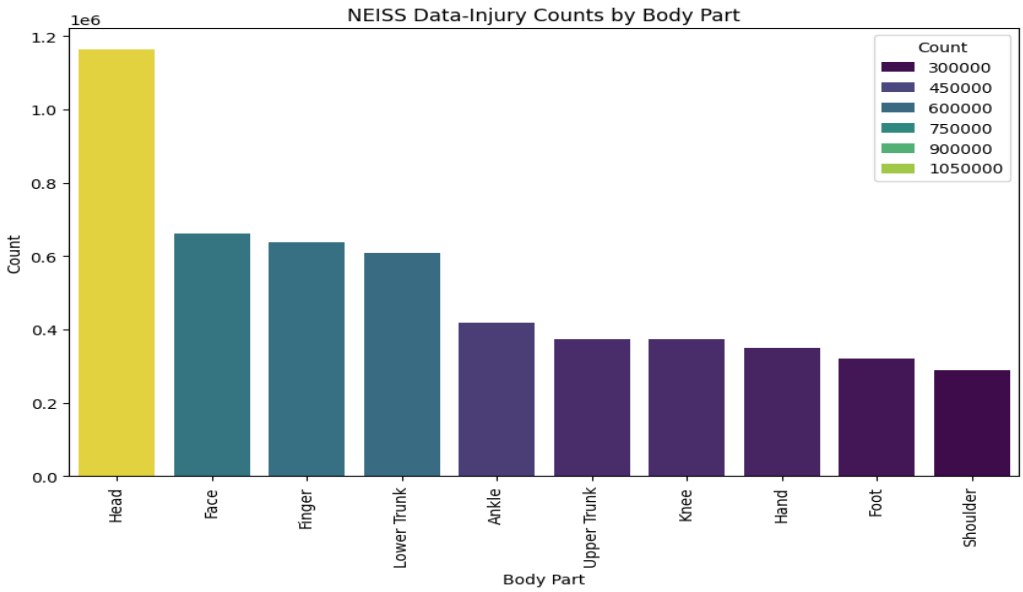
Whites accounted to the most of the injuries' reported into the ER department.

Injury Count by Other Race



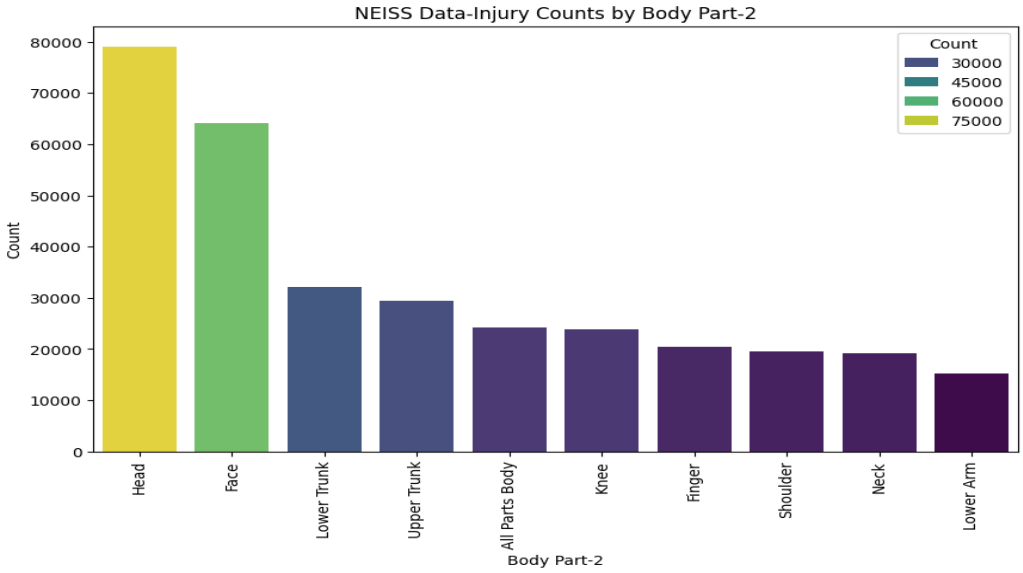
Hispanics accounted for the highest number of injuries reported.

Injury by Body Part



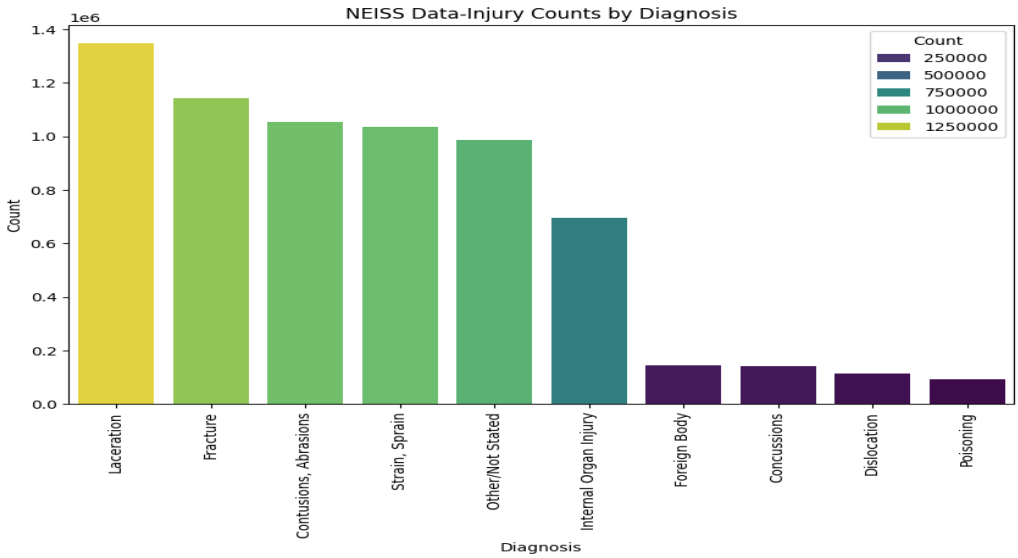
Head followed by Face, Finger and Lower Trunk contributed to the highest number of injuries.

Injury by Body Part 2



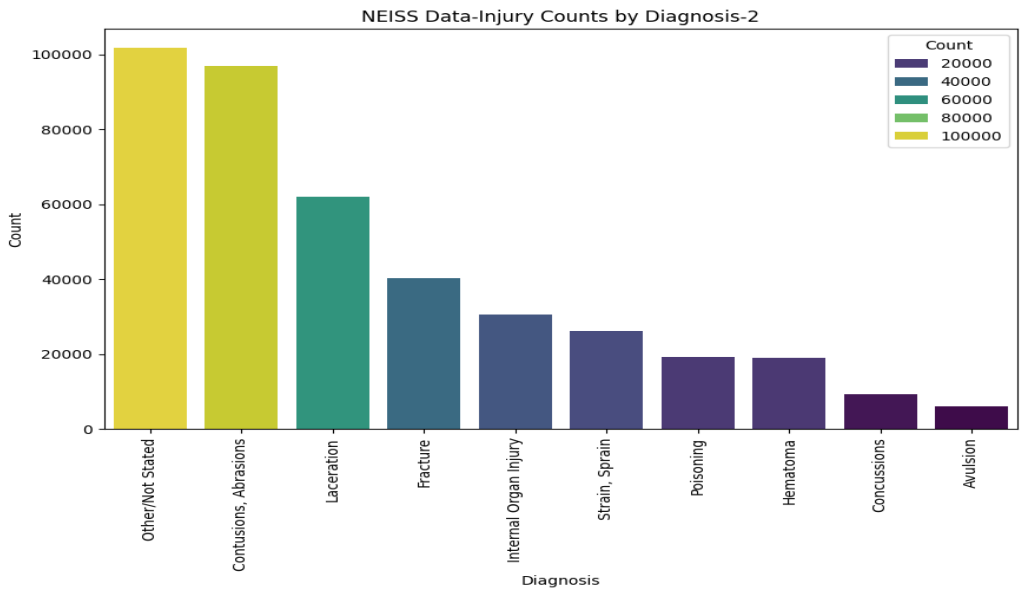
Head followed by Face and Lower/Upper Trunk contributed to the highest number of injuries.

Injury count by diagnosis



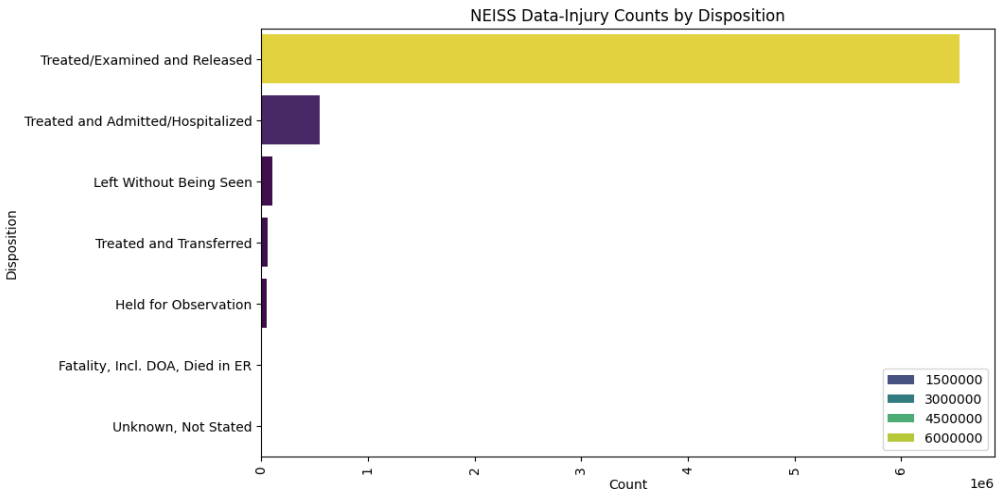
Laceration followed by Fracture, Abrasions and Strain/Sprain contributed to the highest number of injuries.

Injury Count by Diagnosis 2



Abrasions followed by Lacerations and Fracture contributed to the highest number of injuries. We also see a huge number of observations without a Diagnosis 2. This could be from pre 2019 data or could be because of the absence of secondary diagnosis.

Injury Count by Disposition



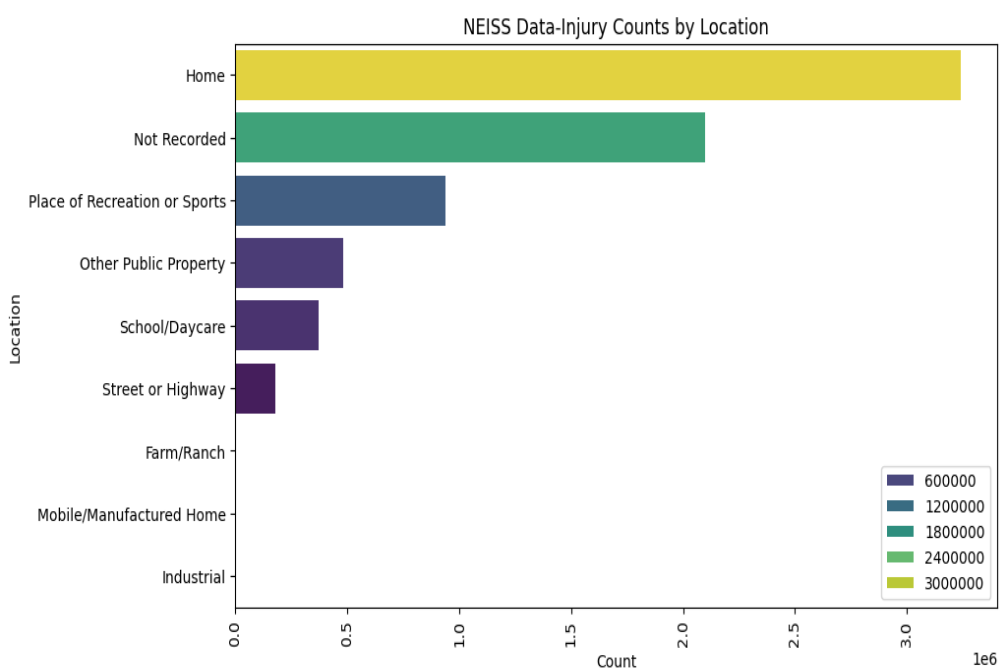
Most of the patients who went to the ER were Treated and Released. The NEISS data seems to be unbalanced, and we may need to use techniques to over sample the minority class. 90% of all the Injuries were treated/examined and released. 7.5% of the Injuries were Admitted/Hospitalized.

Disposition_Code	Disposition_Description	Proportion
1	Treated/Examined and Released	0.90
4	Treated and Admitted/Hospitalized	0.08
6	Left Without Being Seen	0.01
2	Treated and Transferred	0.01
5	Held for Observation	0.01
8	Fatality, Incl. DOA, Died in ER	0.00
9	Unknown, Not Stated	0.00

Based on the disposition, we need to create a new feature 'Hospitalized' that groups the below dispositions.

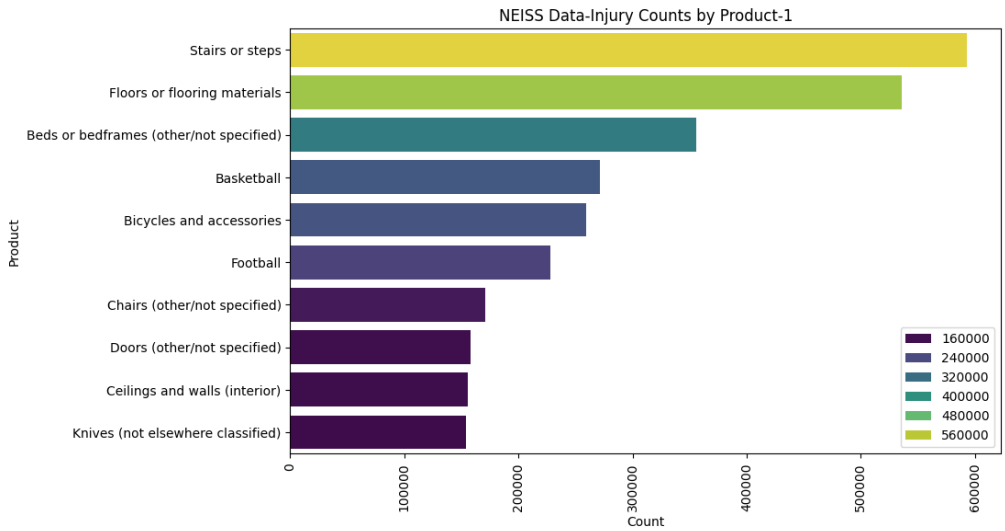
Disposition Code	Hospitalized (Yes/No)
4,2,5,8	Yes
1,6,9	No

Injury Count by Location of Injury



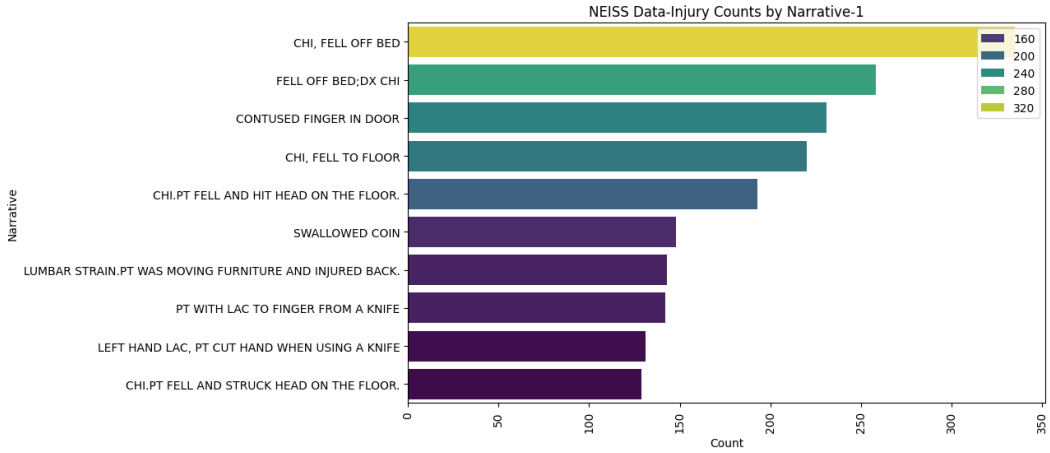
Most of the injuries happened at home followed by not recorded/unknown, sports, or at public property.

Injury by Product



Stairs/Steps followed by Floors and Beds seems to be causing the highest injuries. Recreational activities/Sports seems to be also causing lot of injuries.

Injury by Narrative



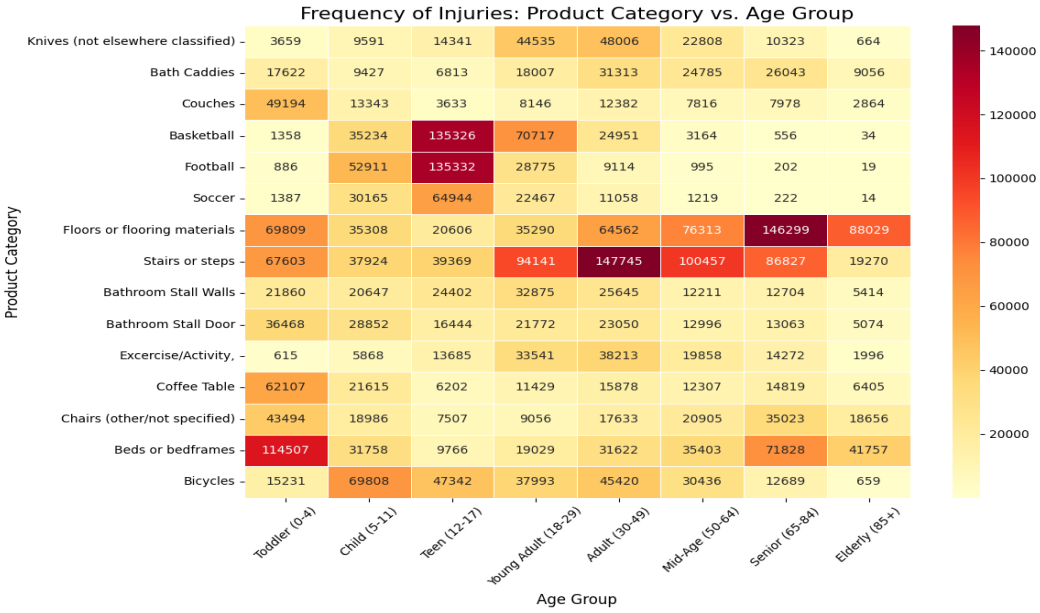
Most of the injury seems to have happened by Falling of the bed or by contusion of finger in door.

Word Cloud of Narrative

The word cloud from the narrative highlights falls, head, laceration, finger, injury during play as most prominent words.

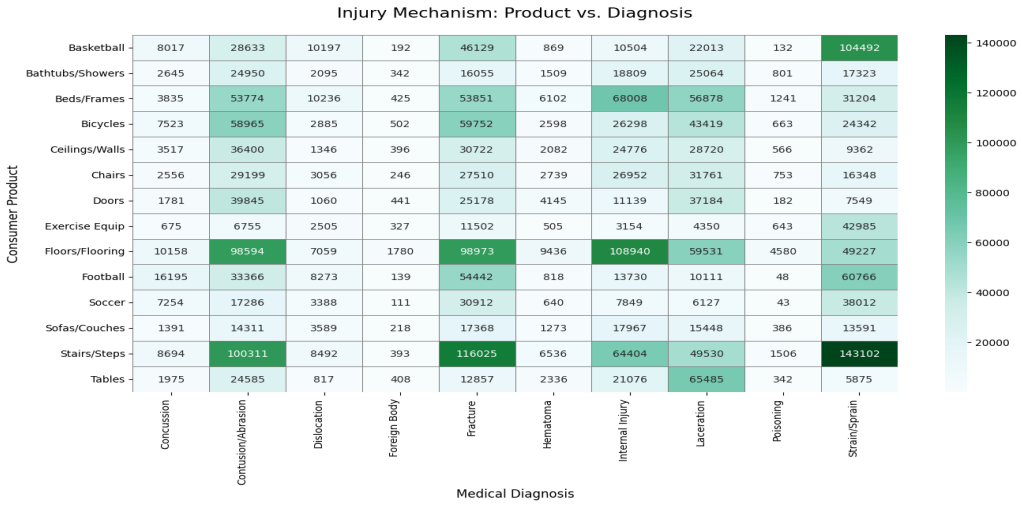
2.3.2 Bivariate Relationships

Product Categories Vs Age Group

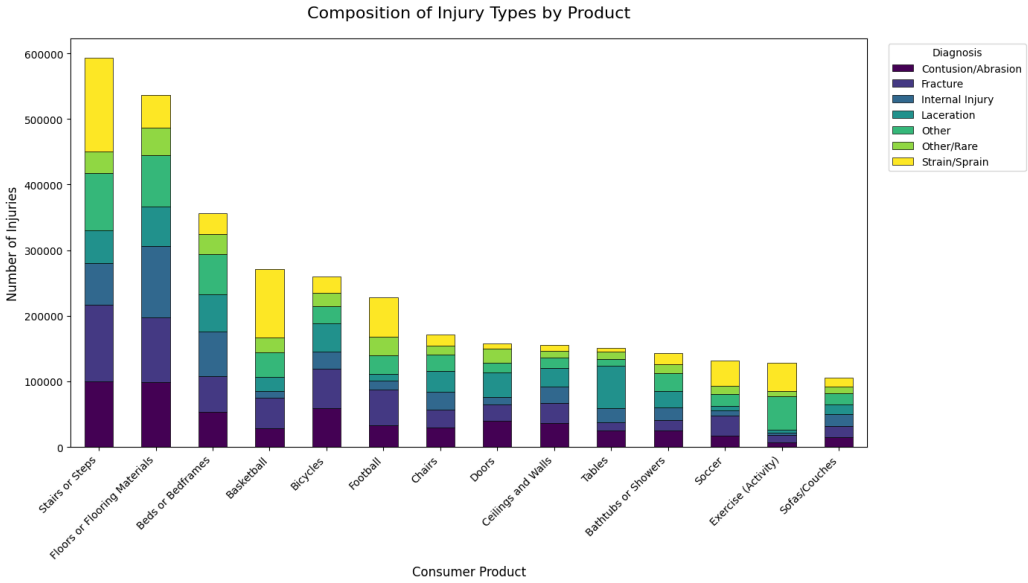


The above charts list the product categories that is dangerous across age groups.

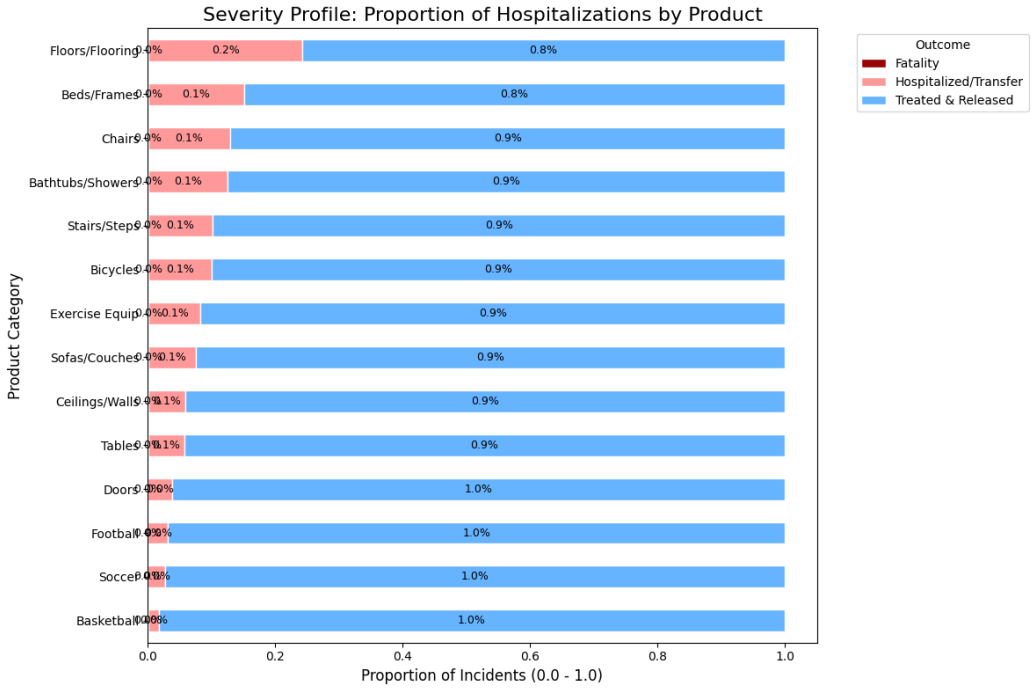
Product Categories Vs Diagnosis



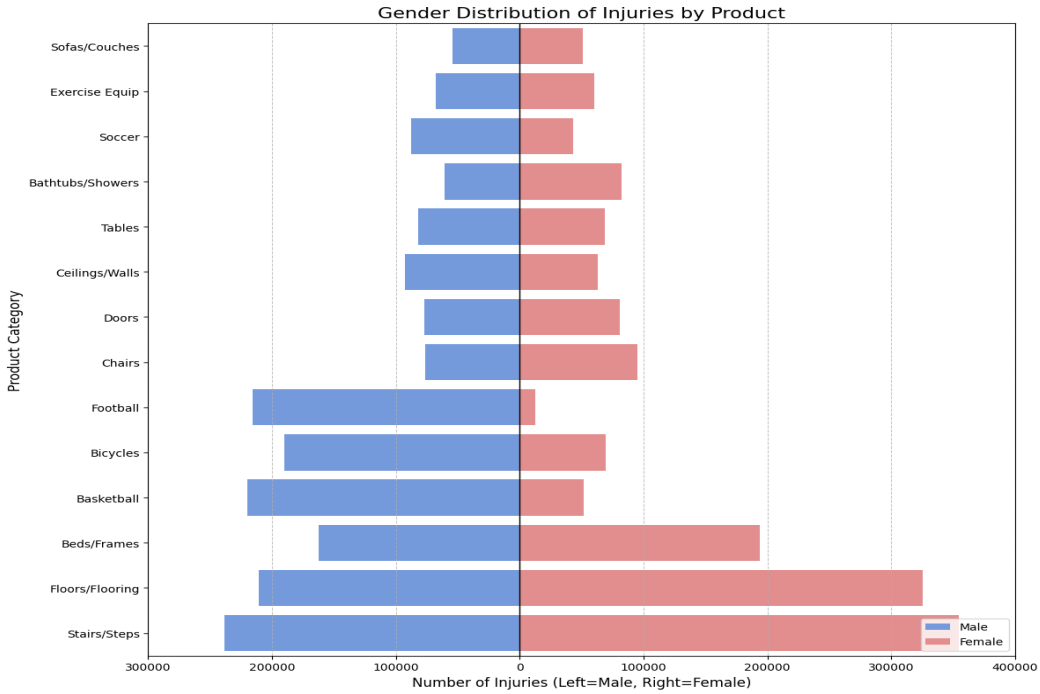
The above charts list the product categories and the corresponding injuries caused by the usage.



Disposition Vs Product

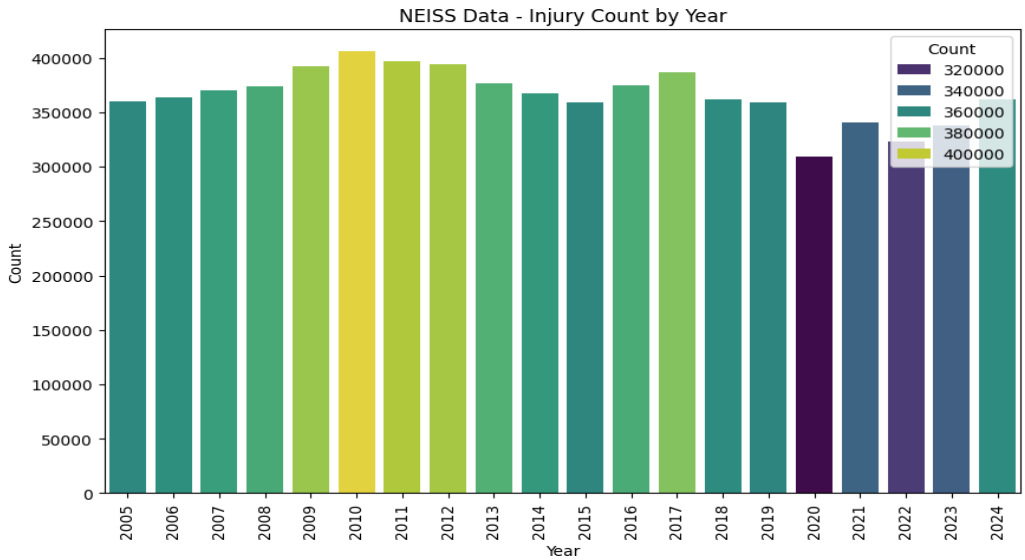


Sex Vs Product

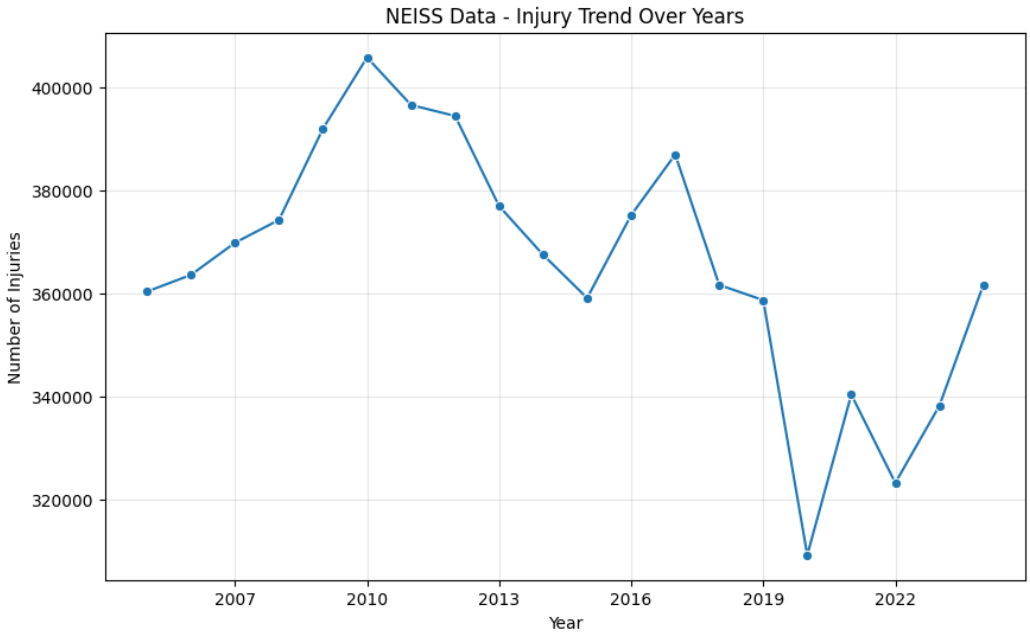


2.3.3 Temporal Patterns and Trends

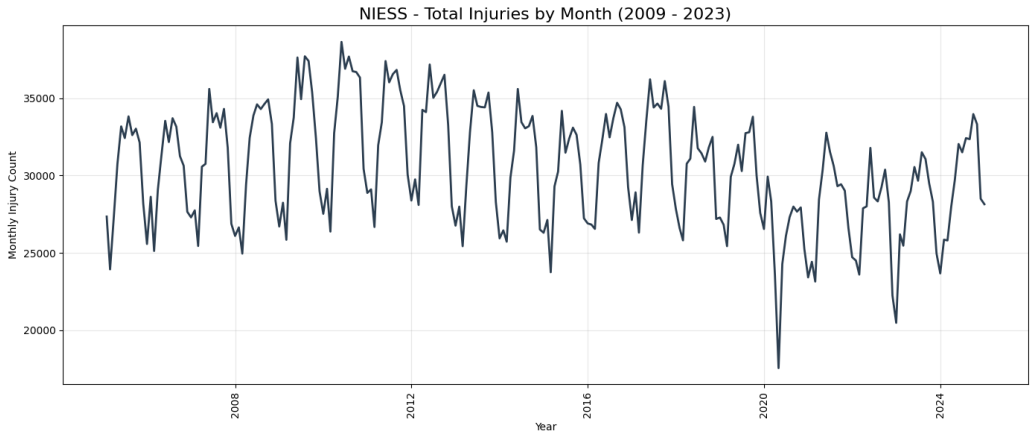
Injury Count over year



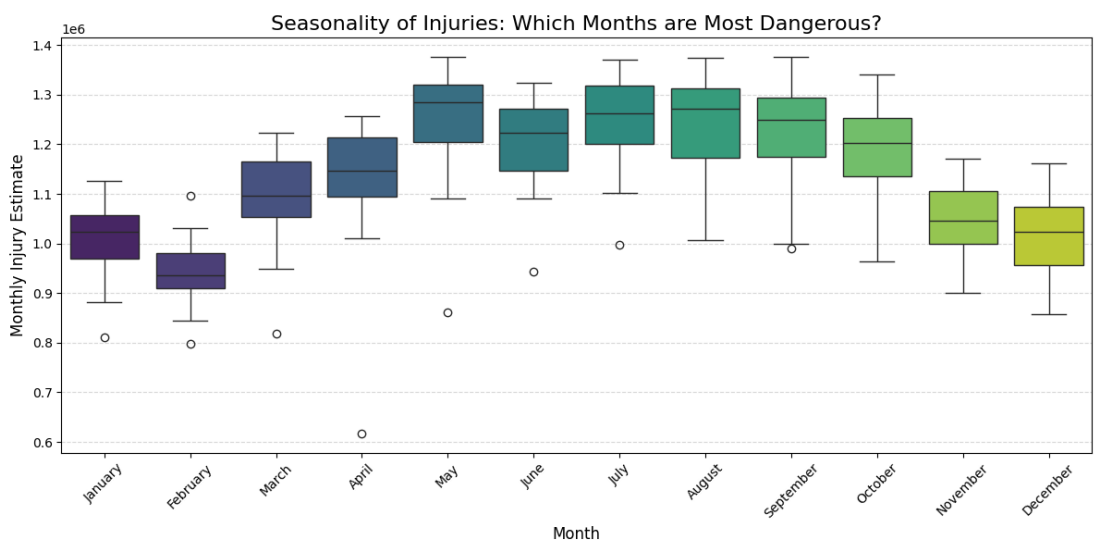
Injury Trend over years



Count of Injury Data by Month



Seasonal Component - We see a seasonal component in the data for every year. The number of injuries peaks during the summer months and reduces in fall and spring. This could be possibly due to the reduction in mobility and social activities during these months. The only exception to this is seen in the year 2019-2020 where the overall injury itself has reduced due to an external factor - the COVID 2019 factor. After the pandemic, the rate of injuries has slowly picked up and follows the seasonal trend like before. This shows the impact of external factors/shocks like the pandemic or mobility/economic (fewer people involved in recreational, work or public activities) reasons causing this decrement. The seasonal factor is robust and is predictable. The data exhibits a seasonal cyclicality and declining trend between 2005 to 2011 shows an upward trend in the injury counts where in 2011 it tapers off at 400K. The injury counts show a slight decline between 2012 -2019 - this could be again due to an external factor like population or increase in the safety awareness contributing to the slight decline.

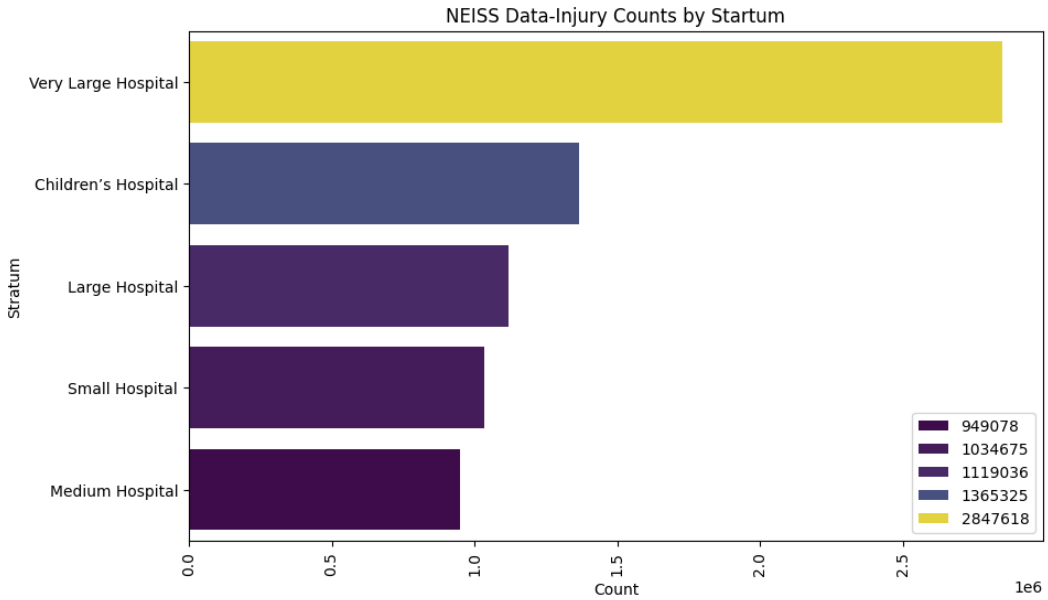


The data points to three critical insights:

1. Injury rates are heavily dependent on population mobility and economic activity
2. Seasonal factors remain robust and predictable, and
3. Exogenous shocks (pandemic, policy changes) can produce sustained structural shifts in injury epidemiology. For public health planning and emergency preparedness, the pre-2020 baseline may no longer reflect current injury risk, and forward projections should account for the new post-pandemic normal.
4. Summary months have the most injuries, and we see a fall in the injuries in the spring and fall.

2.3.4 Organizational Level Segmentation

Injury by Hospital Stratum



A lot of injuries seem to have happened in large hospitals.

2.4 Data Quality Assessment

2.4.1 Completeness Analysis

Missing/Truncated Data Quantification and Analysis

Summary of features with NULL or Missing Data with percentage data missing and NULL

	Column Name	Null Count	Null %	Unknown Count	Total_Missing(Null & Unknown)	Percentage Missing
16	Fire_Involvement	0	0.00	7283693	7283693	99.56
13	Other_Diagnosis_2	7214095	98.61	0	7214095	98.61
12	Diagnosis_2	6866693	93.86	0	6866693	93.86
11	Body_Part_2	6866693	93.86	0	6866693	93.86
6	Other_Race	6821041	93.24	0	6821041	93.24
10	Other_Diagnosis	6329255	86.52	0	6329255	86.52
20	Alcohol	5283928	72.23	0	5283928	72.23
21	Drug	5283928	72.23	0	5283928	72.23
7	Hispanic	5283928	72.23	0	5283928	72.23
5	Race	0	0.00	2145028	2145028	29.32
15	Location	0	0.00	2099509	2099509	28.70
4	Sex	0	0.00	978	978	0.01
14	Disposition	0	0.00	382	382	0.01

Fire Involvement: A fire Involvement with a value of 0 could mean that either no fire was involved or fire involvement not recorded. 99.56% of the observations are missing this information. We can scan the narrative attribute to see if can identify a Fire involvement – if not, we might need to ignore this attribute.

Other Diagnosis_2: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. This is the field to store additional diagnosis if needed. 98.61% of the data is missing this information. We might need to ignore this feature.

Diagnosis_2: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. This is the field to store additional diagnosis if needed. 93.86% of the data is missing this feature. We might need to ignore this feature.

Body Part_2: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. This is the field to store additional information of the body part that was injured. 93.86% of the data is missing this information and we might need to ignore this feature.

Other Race: This attribute is only used we have a “3- Other” entered in the race attribute. We need to use this as a separate category. 93.24% of the data is missing this information.

Other Diagnosis: This feature is only used we have a “71 – Other/Not Stated” entered in the diagnosis attribute. 86.52% of the data is missing this information.

Alcohol: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. We need to use this attribute in combination with the Narrative feature to check if there is a true alcohol usage involved when the injury occurred.

Drug: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. We need to use this attribute in combination with the Narrative feature to check if there is a true Drug usage involved when the injury occurred.

Hispanic: This feature was added for data beyond 2019.This could be one reason why the feature has many unknown values. We need to use this attribute in combination with the Narrative feature to check if there is a true Drug usage involved when the injury occurred.

Sex: 978 records are missing this information and contributes to the 0.01% of the overall data. We can update the majority class information for this missing data, and this will not impact the modelling.

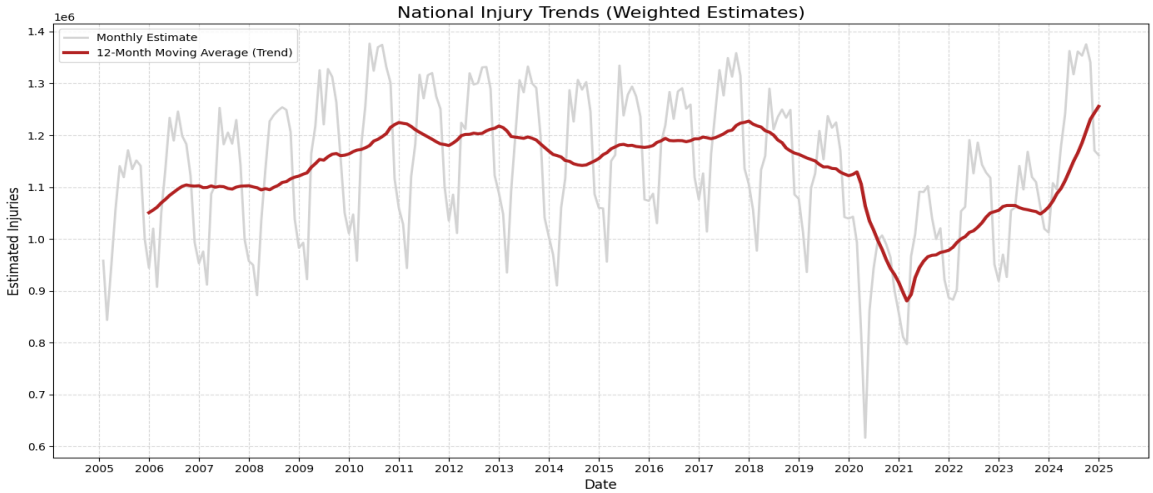
Race: 29% of the records does not have a race or has a 0-not/stated in ED record. We will treat this as a separate category during modelling.

Disposition: 382 records have a disposition value ‘9 – Unknown/Not Stated’. During the feature engineering phase, we may need to consider this as a category that does not cause Hospitalization.

Field Truncation: The data in many of the narrative field is truncated and is un-usable.

Completeness Trend Overtime

The NEISS data from January 2005 to December 2024 is complete and the dataset has observation for each day.



2.4.2 Accuracy Assessment

None of the data from the year January of 2005 – December 2024 is duplicated and each record corresponds to a unique ER admission

The data in NEISS dataset is from the medical records that are entered into the system at high stress condition and hence prone to errors (like hospitals misclassifying the products)

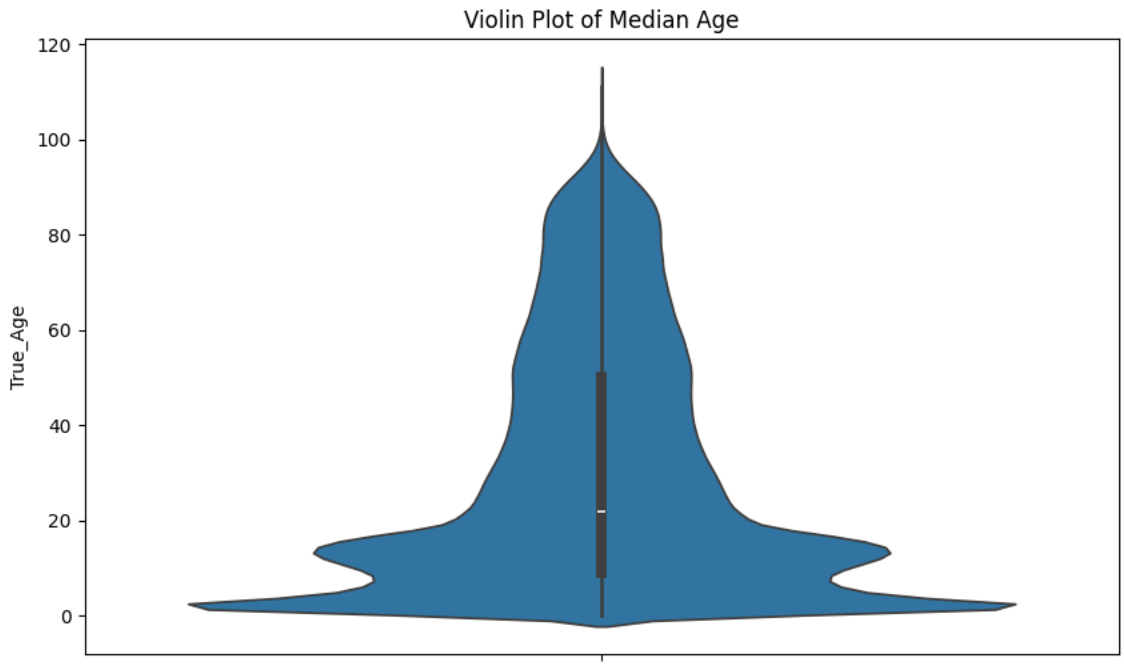
The data sampling is taken from approximately 100 hospitals and is weighted to represent the entire population in the USA. This could possibility induce selection bias as NEISS only incudes injuries reported at the ER departments and ignores, the ones that are self-treated, treated by the primary care physicians or at the urgent care. We need to use weight in conjugation to the stratum to generate statistically valid national estimates.

Range Validation of Numerical Fields:

Age:

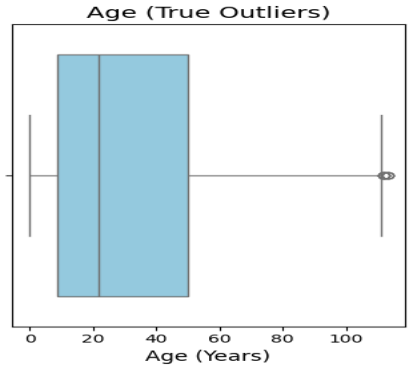
The age falls in plausible range from 0.8 – 113 Years.

```
count   7315732.00
mean         30.95
std          26.16
min           0.08
25%           9.00
50%          22.00
75%          50.00
max         113.00
```



Most of the data is of patients between 0.8-40 years of age.

Outlier Detection for Age:



Treatment Date:
The range of treatment date is from 01/01/2005 to 31/12/2024 and does not fall outside of this range.

2.4.3 Consistency Evaluation

Standardization Issues:
Age Attribute:
NEISS uses unique code to represent ages < 2 years, and we need to make sure that we convert age to a common scale of years. After the conversion, if there are any records with an age of zero, we can either drop them if the volume is low or we can use the median age to substitute the data.

Temporal Consistency:
NEISS product codes are susceptible to change and NEISS could add, delete or merge codes annually and this could create inconsistencies in the product codes. The below list of products 47 products shows inconsistent reporting and does not show up in some years.
Product Codes that are inconsistent:
394, 556, 569, 574, 876, 897, 899, 1132, 1145, 1317, 1329, 1381, 1394, 1395, 1417, 1465, 1466, 1527, 1543, 1545, 1552, 1606, 1618, 1620, 1629, 1706, 1707, 1710, 1841, 1855, 1909, 3215, 4802, 5004, 5016, 5017, 5019, 5020, 5021, 5022, 5023, 5024, 5025, 5042, 5043, 5045, 5046
Contingency: We could remove records from the dataset with these products.

Sampling Consistency:
The number of participating hospitals in NEISS program could vary every year. The weight variable adjusts the for this variation. We should always use the weight variable when comparing the data across years and using raw count to compare the data across years is statistically invalid.

2.4.4 Timelines and Currency

Data Freshness:
The data is entered into the NEISS system daily by the NEISS hospital coordinator. The finalized data for the prior year is made available by April of the next year. The data is available to perform seasonal and trend analysis. We will be able to perform the analysis of Finalized data till the prior year at any point of time.

Data Updates:
When we query the NEISS database online, we can obtain current year live data, but this data is susceptible to updates. Once the financial year ends, the data is then released for public consumption after cleaning and review, and this data is the finalized data. The transient state updates are made daily by the NEISS coordinator and will be reflecting in the NEISS portal. The fields like Disposition, Product code, Diagnosis are updated by the NEISS coordinator daily into the portal. We should not use transient data from the current financial year for analysis and should only use finalized data to reduce the data reproducibility drift.

Temporal Gaps:
NEISS data from January 2005 – December 20024 does not have any gaps and is a good currency for current analysis and can be used for analyzing the trend and seasonality. The data present the NEISS database is updated daily and allows for seasonal analysis.

Historical Depth:
NEISS data is available online for the last 20 years and we will be able to use the data for historical analysis.

2.4.5 Relevance and Coverage

NEISS is the primary dataset used by the US Consumer Product Safety Commission (CPSC) to issue recalls. Using this data hence simulates the workflow used by the safety regulator. This dataset also connects the product to the consequence/diagnosis and is the right data that could help us pinpoint design deficiencies and consumer impact. However, NEISS records only product involvement and not the causality of injury. We need to understand if the product caused an injury by scanning the narrative data but in many cases could be incomplete.

Population Coverage:
Data collected by NEISS could possibility induce selection bias as NEISS only includes injuries reported at the ER departments and ignores, the ones that are self-treated, treated by the primary care physicians or at the urgent care.

Scenario Representation:
NEISS covers only consumer products and doesn't cover the injures caused by products managed by other agencies like Automobiles/Drugs/Medicine/Violence/Work Related Injuries.

Geographic Scope:
The data sampling is taken from approximately 100 hospitals and is weighted to represent the entire population in the USA. This could induce a bias that could miss including local injuries/patterns occurring in areas not represented by these selected hospitals.

2.4.6 Data Quality and Impact Assessment

1. Data Completeness

   Around 30% of the collected data is missing Race (Unknown/Not Recorded) information. This could mean that we will not be able to reliably analyze if a product specifically injures a specific ethnic group. We may instead use the Age and Sex for demographic analysis and not use Race.

   The narrative field of many of the records are truncated and we might not be able to identify the root cause of the injury.

2. Data Accuracy
   NEISS relies on data entered by hospital staff in high stress environments and hence prone to errors. The selection bias systematically underestimates minor injuries (that needs hospitalization) and fatalities.
3. Data Consistency
   Product definitions change over time as CPSC could add, delete, merge products every year. The hospitals involved in the NEISS data also change and hence counting the raw cases year to year is incorrect and we need to use weighted estimates.
4. Validity
   NEISS records only product involvement and not the causality of injury. We need to understand if the product caused an injury by scanning the narrative data but in many cases could be incomplete.